

Graduation Project

ONL4-DAT1-G3



About the
Project

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E-COMMERCE SALES & CUSTOMER BEHAVIOR ANALYSIS



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Project

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UNDER SUPERVISION OF:
PROF. DR. AHMED ISMAIL

GROUP #2

Bola Mourise

Amany Aly

Al Zahraa Samir

LEADER: Asmaa Ismail



Project Overview

Analyze customer purchasing behavior and sales performance.



Data collected from United Kingdom, Germany, Canada, and Australia.



Covers transactions from 2023–2024.



Includes customer demographics:
Age, Gender, Income Level, and Segmentation.



Introduction



High Data Volume Challenge:

Discovery: The dataset was exceptionally large, containing **294,438 rows** of transactional records, which required optimized processing to avoid system lag during analysis.

Action Taken: Utilized Power Query's efficient indexing and data loading capabilities to handle the massive volume smoothly, ensuring performance stability while building visualizations.

294,438

Rows



Introduction



Objectives

Time-Series Analysis: Evaluate sales trends, seasonality, and diagnose the 2023-2024 revenue drop using YoY metrics.

Customer Segmentation: Identify high-value segments (New, Regular, Premium) based on purchase frequency and total spend.

Demographic & Geographic Mapping: Analyze how age, gender, income, and location drive purchasing behavior.

Actionable Recommendations: Deliver interactive dashboards to optimize marketing campaigns and business strategies.

Data Cleaning



Using Excel-Power Query to clean, transform, and handle data anomalies, replacing errors to ensure data integrity and readiness for analysis.

TransactionID	Customer_ID	Customer_Name	Email	Phone	Address	City	State	Zipcode	Country	Age	Gender	Income	Customer_Type	Date	Year	Month	Time	Total_Purchase_Amount	Total_Amount	Product_Category	Product_Name	
8691788	37249	Michelle Hill	Ebony39@gmail.com	1.4E+09	3959 Amalfi	Dortmund	Berlin	77985	Germany	21	Male	Low	Regular	#####	2023	September	22:03:55	3	108.029	324.086	Clothing	Nike
2174773	69749	Kelsey Hill	Mark36@gmail.com	6.9E+09	82072 Dav	Nottingham	England	99071	UK	19	Female	Low	Premium	#####	2023	December	8:42:04	2	403.354	806.708	Electronics	Samsung
6679610	30192	Scott Jensen	Shane85@gmail.com	8.4E+09	4133 Your	Geelong	New South	75929	Australia	48	Male	Low	Regular	#####	2023	April	4:06:29	3	354.478	1063.43	Books	Penguin
7232460	62101	Joseph Miller	Mary34@gmail.com	2.8E+09	8148 Thor	Edmonton	Ontario	88420	Canada	56	Male	High	Premium	5/8/2023	2023	May	14:55:17	7	352.408	2466.85	Home Decor	Home Depot
4983775	27901	Debra Cole	Charles30@gmail.com	9.1E+09	5813 Lori	Bristol	England	48704	UK	22	Male	Low	Premium	#####	2024	January	16:54:07	2	124.277	248.553	Grocery	Nestle
6095326	41289	Ryan Johnson	Haley12@gmail.com	3.3E+09	532 Ashley	Brisbane	New South	74430	Australia	58	Female	Medium	Premium	#####	2023	September	23:24:27	4	296.292	1185.17	Electronics	Apple
5434096	97285	Erin Lewis	Arthur76@gmail.com	1.6E+09	600 Brian	Kitchener	Ontario	47545	Canada	29	Female	Low	New	#####	2023	June	13:35:51	2	315.058	630.115	Electronics	Samsung
2344675	26603	Angela Fiel	Tanya94@gmail.com	3.7E+09	237 Young	Munich	Berlin	86862	Germany	29	Male	Medium	Premium	#####	2023	March	10:12:56	1	46.5881	46.5881	Clothing	Zara
4155845	80175	Diane Clark	Martin39@gmail.com	6.2E+09	8823 Maria	Wollongong	New South	39820	Australia	46	Male	Medium	New	1/6/2024	2024	January	14:38:26	8	328.839	2630.71	Grocery	Nestle
4926148	31878	Lori Bell	Jessica33@gmail.com	6E+09	6225 Willi	Cologne	Berlin	64317	Germany	25	Male	Medium	New	#####	2023	October	22:27:40	10	397.611	3976.11	Home Decor	Home Depot
8493213	19136	Jonathan E	Mark38@gmail.com	3E+09	9772 Sosa	Portsmouth	England	59280	UK	64	Female	Low	Regular	#####	2023	July	23:06:51	4	90.9819	363.927	Home Decor	Home Depot
1609659	66883	Brianna O'Connell	David47@gmail.com	9.4E+09	95471 Jerr	Portsmouth	England	91253	UK	31	Female	Medium	Regular	#####	2023	June	13:09:58	1	364.831	364.831	Books	Random House
2846832	31930	Kristine Williams	Charles29@gmail.com	1.8E+09	60629 Jon	Portsmouth	England	38173	UK	56	Female	Medium	Regular	1/2/2024	2024	January	0:00:47	10	161.879	1618.79	Grocery	Coca-Cola
3217461	74671	Felicia Brown	Laura72@gmail.com	8.9E+09	381 Richard	Portsmouth	England	32061	UK	53	Female	Medium	Regular	5/7/2023	2023	May	18:49:35	9	350.817	3157.35	Grocery	Nestle
2401331	98300	Andrew Guler	Eric76@gmail.com	2.9E+09	470 Rodrig	Portsmouth	England	5259	UK	32	Female	High	Regular	#####	2023	November	23:41:05	6	297.726	1786.36	Clothing	Adidas
4969987	64995	Justin Lara	Jacob26@gmail.com	2.7E+09	3577 Cole	Portsmouth	England	9845	UK	43	Male	Medium	Regular	#####	2023	June	23:24:22	3	23.7239	71.1717	Books	Random House
6681000	69939	Jessica Harris	Anne45@gmail.com	2.5E+09	93806 Mui	Portsmouth	England	84007	UK	69	Male	Low	Regular	7/1/2023	2023	July	11:12:02	10	14.7272	147.272	Grocery	Pepsi
2430572	71803	Alicia Knight	Jennifer1@gmail.com	6.3E+09	9856 Meye	Portsmouth	England	91967	UK	19	Female	High	Regular	#####	2023	April	5:50:55	3	139.787	419.361	Electronics	Apple
7728099	78376	Ronald Chen	Nicole72@gmail.com	1.8E+09	6048 Char	Portsmouth	England	82975	UK	49	Male	Low	Regular	2/7/2024	2024	February	18:29:59	1	210.138	210.138	Grocery	Coca-Cola
3489336	40654	Sara Rogers	Jeff99@gmail.com	2.7E+09	6048 Char	Portsmouth	England	3035	UK	69	Male	Low	Regular	#####	2023	October	13:01:04	2	469.66	939.319	Home Decor	IKEA
6925572	42165	Daniel Hutcheson	Oscar54@gmail.com	9.6E+09	37402 Wil	Portsmouth	England	29119	UK	46	Male	Low	Regular	#####	2023	March	11:51:41	6	420.875	2525.25	Grocery	Nestle
8871554	63914	Joshua Noe	Diana25@gmail.com	5.4E+09	7467 Henry	Portsmouth	England	16385	UK	61	Male	High	Regular	8/6/2023	2023	August	5:13:31	4	200.287	801.149	Books	HarperCollins

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6679610	30192	Scott Jensen	Shane85@gmail.com	8.4E+09	4133 Your	Geelong	New South Wales	75929	Australia	48	Male	Low	Regular	#####	2023	April	4:06:29	3	354.478	1063.43	Books	Penguin
7232460	62101	Joseph Miller	Mary34@gmail.com	2.8E+09	8148 Thorndike	Edmonton	Ontario	88420	Canada	56	Male	High	Premium	5/8/2023	2023	May	14:55:17	7	352.408	2466.85	Home Decor	Home Depot
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6095326	41289	Ryan Johns	Haley12@gmail.com	3.3E+09	532 Ashley	Brisbane	New South Wales	74430	Australia	58	Female	Medium	Premium	#####	2023	September	23:24:27	4	296.292	1185.17	Electronics	Apple
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4155845	80175	Diane Clark	Martin39@gmail.com	6.2E+09	8823 Maria	Wollongong	New South Wales	39820	Australia	46	Male	Medium	New	1/6/2024	2024	January	14:38:26	8	328.839	2630.71	Grocery	Nestle
4926148	31878	Lori Bell	Jessica33@gmail.com	6E+09	6225 Willis	Cologne	Berlin	64317	Germany	25	Male	Medium	New	#####	2023	October	22:27:40	10	397.611	3976.11	Home Decor	Home Depot
8493213	19136	Jonathan Edwards	Mark38@gmail.com	3E+09	9772 Sosa	Portsmouth	England	59280	UK	64	Female	Low	Regular	#####	2023	July	23:06:51	4	90.9819	363.927	Home Decor	Home Depot
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8871554	63914	Joshua Noe	Diana25@gmail.com	5.4E+09	7467 Henry	Portsmouth	England	16385	UK	61	Male	High	Regular	8/6/2023	2023	August	5:13:31	4	200.287	801.149	Books	HarperCollins

Introduction



Extensive Data Gaps & Missing Values:

Discovery: Discovered widespread missing data (**blanks**) across multiple critical columns. Most were found in the **Date, Year, and Month** columns, as well as the **purchases (Order Q) and Total Amount** columns.

Action Taken: Applied advanced logical formulas to reconstruct the missing timeline elements. Similarly, blank fields in order counts and total amounts were cross-calculated and filled using mathematical dependencies to achieve 100% data completeness and accuracy.

Introduction



Microsoft Excel interface showing a data table and a function argument dialog box.

Excel Ribbon: File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Automate, Help, Power Pivot, Table Design, Query.

Formula Bar: P3923, $=DATE([@Year],[@Year],1)$

Table Data:

	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
	Gender	Income	Customer_Segment	Date	Year	Month	Time	Total_Purchases	Amount	Total_Amount	Product_Category	Product_Brand	Product_Type	Feedback
3923	Female	Medium	Regular	@Year,1	2023	November	6:14:15 PM	1	292.0317	292.0317	Home Decor	Home Depot	Furniture	Average
4344	Male	Medium	Regular		2023	July	3:05:25 AM	7	338.7156	2371.009	Books	HarperCollins	Non-Fiction	Bad
5137	Female	Medium	Regular		2023	July	2:01:47 PM	7	249.6795	1747.7567	Clothing	Nike	Shorts	Bad
5233	Male							7	302.0738	2114.5166	Home Decor	IKEA	Lighting	Bad
6859	Male							5	231.7269	1158.6347	Electronics	Apple	Tablet	Good
7027	Male							2	394.5548	789.1097	Home Decor	Bed Bath & Beyond	Bathroom	Average
7165	Male							4	374.1295	1496.5182	Clothing	Zara	Shirt	Bad
7593	Female							2	214.6814	429.3628	Grocery	Nestle	Snacks	Bad
10339	Male							1	309.4436	309.4436	Clothing	Nike	Shorts	Excellent
11660	Male							10	339.8549	3398.5487	Grocery	Coca-Cola	Soft Drink	Good
11907	Male							10	203.6484	2036.4837	Clothing	Nike	T-shirt	Average
12254	Male							1	259.7956	259.7956	Home Decor	Bed Bath & Beyond	Bedding	Bad
12674	Male							5	33.7423	168.7116	Home Decor	Home Depot	Furniture	Good
13065	Male							1	159.8673	159.8673	Grocery	Coca-Cola	Water	Good
13747	Female							10	468.6886	4686.8858	Home Decor	Bed Bath & Beyond	Bathroom	Bad
14058	Female							4	292.3765	1169.506	Electronics	Sony	Television	Excellent
15101	Male							7	412.399	2886.7929	Clothing	Adidas	T-shirt	Good
15716	Female							5	121.4016	607.0079	Grocery	Pepsi	Soft Drink	Average
17223	Female	Medium	Regular		2023	September	6:17:05 AM	4	300.5434	1202.1735	Electronics	Sony	Headphones	Excellent
17726	Female	Medium	New		2023	April	11:42:50 PM	4	460.3682	1841.4728	Books	Penguin Books	Children's	Average
18226	Female	Medium	New		2023	April	12:04:58 PM	3	13.0011	39.0033	Grocery	Coca-Cola	Juice	Excellent
18601	Male	Medium	New		2023	November	11:52:55 AM	5	161.4753	807.3763	Electronics	Samsung	Television	Excellent
10811	Female	High	New		2023	July	6:57:11 PM	8	398.0436	3184.3484	Home Decor	Bed Bath & Beyond	Bathroom	Good
11060	Female	High	New		2023	October	11:11:03 AM	5	217.0791	1085.3955	Electronics	Samsung	Television	Bad
11549	Male	Low	New		2023	August	3:50:22 PM	1	177.2755	177.2755	Books	HarperCollins	Thriller	Good
11619	Female	Low	New		2024	February	1:06:00 PM	6	201.2545	1207.5269	Clothing	Adidas	T-shirt	Bad

Function Arguments Dialog Box:

DATE

Year: $[@Year]$ = 2023

Month: $[@Year]$ = 2023

Day: 1 = 1

Formula result = 7/1/2191 0:00

Help on this function

OK Cancel

Introduction



AutoSave Off

new_retail_data 1 Ask (version 1).xlsx

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelpPower PivotTable DesignQuery

ClipboardFontAlignmentNumberStylesCellsEditingAdd-ins

CommentsShareCatch up

AutoSumFillClearSort & FilterFind & SelectAdd-ins

02

Introduction



Excel interface showing a spreadsheet with columns: Gender, Income, Customer_Segment, Date, Year, Month, Time, Total_Purchases, Amount, Total_Amount, Product_Category, Product_Brand, Product_Type, Feedback. The formula bar shows `=YEAR([@Date])`.

Function Arguments dialog box for the YEAR function:

- Serial_number: `[@Date]`
- Returns the year of a date, an integer in the range 1900 - 9999.
- Formula result = 2024

Table 1: new_retail_data

	Gender	Income	Customer_Segment	Date	Year	Month	Time	Total_Purchases	Amount	Total_Amount	Product_Category	Product_Brand	Product_Type	Feedback
29	Male	Low	Regular	1/7/2024 0:00	[@Date]	January	5:33:45 PM	5	471.4821	2357.4107	Electronics	Apple	Tablet	Bad
30	Male	Medium	Regular	4/22/2023 0:00		April	9:17:29 AM	8	333.8597	2670.8773	Home Decor	Home Depot	Decorations	Excellent
31	Male	Medium	Regular	6/21/2023 0:00		June	4:42:37 PM	6	240.2282	1441.3691	Grocery	Coca-Cola	Juice	Good
32	Female	Low	Regular	2/6/2024 0:00		February	1:59:20 AM	3	159.8042	479.4126	Clothing	Adidas	Shoes	Good
33	Female	Low	Regular	11/26/2023 0:00		November	6:43:16 AM	2	147.0982	294.1965	Books	HarperCollins	Thriller	Good
34	Male	Medium	Regular	5/22/2023 0:00		May	11:24:20 AM	7	344.6792	2412.7543	Home Decor	IKEA	Decorations	Excellent
35	Male	Low	Regular	6/29/2023 0:00		June					Smartphone			Average
36	Female	Low	Regular	12/16/2023 0:00		December					Chocolate			Excellent
37	Male	Low	Regular	4/10/2023 0:00		April					Decorations			Good
38	Male	High	Regular	12/12/2023 0:00		December					Shirt			Good
39	Male	High	Regular	2/16/2024 0:00		February					Children's Books			Excellent
40	Female	High	Regular	2/8/2024 0:00		February					Soft Drink			Good
41	Male	Low	Regular	10/29/2023 0:00		October					Shoes			Excellent
42	Male	High	Regular	2/21/2024 0:00		February					Juice			Excellent
43	Female	High	Regular	3/17/2023 0:00		March					Literature			Good
44	Female	High	Regular	10/6/2023 0:00		October					Laptop			Bad
45	Male	High	Regular	11/5/2023 0:00		November					Furniture			Excellent
46	Female	High	Regular	12/14/2023 0:00		December					Dress			Average
47	Male	Low	Regular	6/19/2023 0:00		June	11:40:40 AM	1	399.7871	399.7871	Electronics	Sony	Headphones	Good
48	Male	High	Regular	3/16/2023 0:00		March	1:42:10 AM	7	315.6289	2209.4021	Home Decor	IKEA	Lighting	Excellent
49	Female	Medium	Regular	7/16/2023 0:00		July	5:20:37 PM	8	93.0042	744.0338	Home Decor	Home Depot	Decorations	Good
50	Female	Medium	Regular	4/19/2023 0:00		April	2:13:39 AM	7	346.7196	2427.0371	Home Decor	Bed Bath & Beyond	Bedding	Bad
51	Female	Medium	Regular	9/10/2023 0:00		September	7:22:43 PM	6	106.5067	639.0405	Clothing	Adidas	Jacket	Average
52	Male	Low	Regular	3/15/2023 0:00		March	1:07:58 AM	2	230.8759	461.7517	Home Decor	Home Depot	Furniture	Bad
53	Male	High	Regular	11/11/2023 0:00		November	3:38:31 PM	3	394.6063	1183.8188	Grocery	Nestle	Chocolate	Excellent
54	Male	High	Regular	7/25/2023 0:00		July	8:29:30 PM	2	281.6002	563.2003	Electronics	Samsung	Television	Average
55	Female	High	Regular	10/3/2023 0:00		October	4:34:26 AM	4	303.7899	1215.1595	Electronics	Sony	Headphones	Bad

Introduction



Absence of Profit Margin Data:

Discovery: The original dataset lacked a dedicated column for profit margins.

Action Taken: Established a calculated parameter assuming an estimated profit margin of **20% of Total Amount**.

Introduction



= [@[Total_Amount]]*20%									
O	P	Q	R	S	T	U	V	W	X
Date	Year	Month	Time	Total_Purchases	Amount	Total_Amount	Estimated profit (20%)	Product_Category	Product_Brand
10/3/2023 0:00	2023	10	6:31:16 AM	8	\$ 424.62	\$ 3,396.99	\$679.40	Books	HarperCollins
12/24/2023 0:00	2023	12	3:53:12 AM	5	\$ 210.38	\$ 1,051.90	\$210.38	Grocery	Coca-Cola
8/5/2023 0:00	2023	8	10:20:29 PM	4	\$ 289.33	\$ 1,157.31	\$231.46	Home Decor	IKEA
12/3/2023 0:00	2023	12	1:17:50 AM	9	\$ 403.84	\$ 3,634.55	\$726.91	Electronics	Samsung
4/3/2023 0:00	2023	4	11:03:12 AM	6	\$ 227.37	\$ 1,364.23	\$272.85	Grocery	Pepsi
3/18/2023 0:00	2023	3	9:57:49 PM	10	\$ 215.92	\$ 2,159.23	\$431.85	Home Decor	IKEA
7/19/2023 0:00	2023	7	7:39:46 PM	3	\$ 438.61	\$ 1,315.82	\$263.16	Electronics	Sony
12/2/2023 0:00	2023	12	12:29:35 PM	7	\$ 486.13	\$ 3,402.92	\$680.58	Clothing	Zara
9/10/2023 0:00	2023	9	8:58:12 PM	2	\$ 383.27	\$ 766.54	\$153.31	Grocery	Pepsi
3/6/2023 0:00	2023	3	5:35:27 AM	5	\$ 432.69	\$ 2,163.47	\$432.69	Clothing	Zara
12/18/2023 0:00	2023	12	5:39:50 AM	9	\$ 300.89	\$ 2,708.02	\$541.60	Books	HarperCollins
4/6/2023 0:00	2023	4	11:23:34 AM	10	\$ 342.01	\$ 3,420.06	\$684.01	Home Decor	IKEA
10/16/2023 0:00	2023	10	10:36:04 AM	4	\$ 129.46	\$ 517.84	\$103.57	Electronics	Samsung
10/9/2023 0:00	2023	10	6:12:36 AM	4	\$ 361.68	\$ 1,446.72	\$289.34	Electronics	Sony
5/29/2023 0:00	2023	5	9:03:51 AM	8	\$ 418.83	\$ 3,350.65	\$670.13	Grocery	Coca-Cola
1/16/2024 0:00	2024	1	9:10:53 PM	1	\$ 170.99	\$ 170.99	\$34.20	Clothing	Nike
4/28/2023 0:00	2023	4	2:08:16 AM	2	\$ 447.69	\$ 895.37	\$179.07	Electronics	Samsung
3/14/2023 0:00	2023	3	2:44:53 PM	5	\$ 360.40	\$ 1,802.02	\$360.40	Clothing	Nike
9/30/2023 0:00	2023	9	11:52:59 PM	3	\$ 410.28	\$ 1,230.85	\$246.17	Grocery	Pepsi
10/8/2023 0:00	2023	10	11:20:04 PM	10	\$ 195.92	\$ 1,959.17	\$391.83	Electronics	Samsung
9/23/2023 0:00	2023	9	5:44:37 PM	4	\$ 15.73	\$ 62.93	\$12.59	Grocery	Nestle

Data Cleaning



Now, the data is ready to be used in analysis

Transaction_ID	Customer_ID	Name	Email	Phone	Address	City	State	Zipcode	Country	Age	Gender
2096454	86001	Joshua Lee	Jesse17@gmail.com	914618963	76477 Alexander Fall	Calgary	Ontario	66868	Canada	20	Male
7439591	14457	John Baker	Dylan57@gmail.com	224757932	56381 Kevin Mews	St. John's	Ontario	12180	Canada	20	Female
4189818	20896	Sarah Riggs	Ethan10@gmail.com	647181738	2531 Pamela Spring	London	Ontario	63574	Canada	20	Male
1932343	76837	Jimmy Davis	Rachel24@gmail.com	143935093	380 William Trafficway Suite 285	Sheffield	England	6242	UK	20	Female
1924566	29688	David Burke	Cody38@gmail.com	816703885	666 Taylor Rapid	Dallas	Delaware	19892	USA	20	Male
2141993	28913	Madison Mills	Anthony87@gmail.com	289562982	72080 Stephanie Highway Suite 461	Chicago	Connecticut	9514	USA	20	Male
7331794	64037	Courtney James	Melissa7@gmail.com	948774988	99058 Linda Lane Apt. 803	Chicago	Connecticut	72065	USA	20	Male
3356679	94535	Phillip Curtis	Megan74@gmail.com	223909513	27165 Smith Parkways	Chicago	Connecticut	37344	USA	20	Male
5097745	83050	Brianna Manning	Richard1@gmail.com	938758017	9239 Smith Spurs Apt. 165	Chicago	Connecticut	23801	USA	20	Female
9735343	68271	Sarah Weaver	Darrel25@gmail.com	651166816	57003 Wagner Courts Apt. 814	Chicago	Connecticut	35095	USA	40	Female
1990447	64958	Teresa Conrad	Harry97@gmail.com	904457587	2718 Erik Ridges	Chicago	Connecticut	22856	USA	34	Male
3828749	60986	Angela Schwartz	Lisa22@gmail.com	899678886	81206 Armstrong Park	Chicago	Connecticut	36242	USA	34	Female
9881924	53639	Diana Martin	Laura20@gmail.com	433883515	2809 Wood Estates Suite 953	Chicago	Connecticut	40263	USA	34	Female
6573000	78073	Heather Juarez	Travis92@gmail.com	508907541	902 Christian Unions Suite 340	Chicago	Connecticut	72785	USA	34	Female
5581220	29236	Wayne Kemp	Katherine100@gmail.com	252534706	4486 Christopher Roads	Chicago	Connecticut	32361	USA	34	Female
6140966	63569	Jason Ray	Daniel98@gmail.com	486295616	522 Michael Spurs Suite 280	Chicago	Connecticut	27655	USA	34	Female
4313154	94804	Mark Thompson	Mark5@gmail.com	190479779	60192 Bond Trail	Chicago	Connecticut	19715	USA	34	Male
3777308	57217	Barbara Fry	Angela50@gmail.com	500578547	11557 Melanie Island	Chicago	Connecticut	37220	USA	34	Female
8318151	24233	Angela Castro	Tammy5@gmail.com	628502966	6253 Bonnie Drive	Chicago	Connecticut	31066	USA	34	Female
6810146	99539	Javier Pacheco	Adam28@gmail.com	901707733	61240 Keith Viaduct Apt. 087	Chicago	Connecticut	65603	USA	34	Male
2840220	39019	Kelly Turner	Jessica94@gmail.com	819117119	58877 Black Overpass Suite 515	Chicago	Connecticut	90428	USA	34	Male
7569832	33482	Diane Lopez	Alicia7@gmail.com	224982805	02042 Randall Mountains	Chicago	Connecticut	55030	USA	34	Male
1135694	43106	Brian Thompson	Thomas94@gmail.com	792181867	3389 Erica Bypass Apt. 989	Chicago	Connecticut	19774	USA	34	Female
7953987	11472	Jessica Barnett	Melissa20@gmail.com	452507757	427 Clay Lodge	Chicago	Connecticut	34045	USA	34	Male
2988348	30496	Ronald Cole	Don35@gmail.com	172392755	26119 Torres Neck Apt. 060	Chicago	Connecticut	75663	USA	34	Male
5929057	99920	Sandra Young	Jose74@gmail.com	527368430	5095 Anthony Forges	Chicago	Connecticut	30854	USA	34	Male
5997455	23605	James Jones	Edward89@gmail.com	821415860	1421 Johnny Overpass Apt. 297	Chicago	Connecticut	47902	USA	34	Female
6473464	41019	Pamela Franklin	Matthew1@gmail.com	507639012	4983 Meyer Parkway	Chicago	Connecticut	92736	USA	34	Male
6096537	26139	Edward Gonzales	Andrea69@gmail.com	574055012	7781 Thomas Isle Suite 164	Chicago	Connecticut	59414	USA	34	Female
2521057	78137	Jordan Cook	Michelle76@gmail.com	582841341	895 Devon Summit	Chicago	Connecticut	73081	USA	34	Male

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2521057	78137	Jordan Cook	Michelle76@gmail.com	582841341	895 Devon Summit	Chicago	Connecticut	73081	USA	34	Male

ent	 Date	 Year	 Month	 Time	 Total_Purchases	 Amount	 Total_Amount	 Product_Category	 Product_Brand	 Product_Type	 Feedback	 Shipping_Method	 Payment_Method	 Order_Status
	10/3/2023 0:00	2023	10	6:31:16 AM	8	\$ 424.62	\$ 3,396.99	Books	HarperCollins	Fiction	Excellent	Express	PayPal	Processing
	12/24/2023 0:00	2023	12	3:53:12 AM	5	\$ 210.38	\$ 1,051.90	Grocery	Coca-Cola	Juice	Good	Same-Day	Cash	Shipped
	8/5/2023 0:00	2023	8	#####	4	\$ 289.33	\$ 1,157.31	Home Decor	IKEA	Furniture	Bad	Same-Day	Debit Card	Processing
	12/3/2023 0:00	2023	12	1:17:50 AM	9	\$ 403.84	\$ 3,634.55	Electronics	Samsung	Television	Average	Express	Cash	Processing
	4/3/2023 0:00	2023	4	#####	6	\$ 227.37	\$ 1,364.23	Grocery	Pepsi	Soft Drink	Excellent	Express	PayPal	Processing
	3/18/2023 0:00	2023	3	9:57:49 PM	10	\$ 215.92	\$ 2,159.23	Home Decor	IKEA	Furniture	Bad	Same-Day	Debit Card	Shipped
	7/19/2023 0:00	2023	7	7:39:46 PM	3	\$ 438.61	\$ 1,315.82	Electronics	Sony	Television	Bad	Same-Day	Credit Card	Pending
	12/2/2023 0:00	2023	12	#####	7	\$ 486.13	\$ 3,402.92	Clothing	Zara	Jeans	Excellent	Express	Cash	Shipped
	9/10/2023 0:00	2023	9	8:58:12 PM	2	\$ 383.27	\$ 766.54	Grocery	Pepsi	Water	Average	Standard	Debit Card	Pending
	3/6/2023 0:00	2023	3	5:35:27 AM	5	\$ 432.69	\$ 2,163.47	Clothing	Zara	Dress	Good	Same-Day	Credit Card	Pending
	12/18/2023 0:00	2023	12	5:39:50 AM	9	\$ 300.89	\$ 2,708.02	Books	HarperCollins	Thriller	Good	Standard	Cash	Shipped
	4/6/2023 0:00	2023	4	#####	10	\$ 342.01	\$ 3,420.06	Home Decor	IKEA	Furniture	Good	Unknown	Debit Card	Delivered
	10/16/2023 0:00	2023	10	#####	4	\$ 129.46	\$ 517.84	Electronics	Samsung	Television	Excellent	Express	PayPal	Shipped
	10/9/2023 0:00	2023	10	6:12:36 AM	4	\$ 361.68	\$ 1,446.72	Electronics	Sony	Television	Excellent	Standard	PayPal	Shipped
	5/29/2023 0:00	2023	5	9:03:51 AM	8	\$ 418.83	\$ 3,350.65	Grocery	Coca-Cola	Soft Drink	Bad	Same-Day	Cash	Delivered
	1/16/2024 0:00	2024	1	9:10:53 PM	1	\$ 170.99	\$ 170.99	Clothing	Nike	Shorts	Good	Express	Credit Card	Processing
	4/28/2023 0:00	2023	4	2:08:16 AM	2	\$ 447.69	\$ 895.37	Electronics	Samsung	Tablet	Bad	Express	PayPal	Processing
	3/14/2023 0:00	2023	3	2:44:53 PM	5	\$ 360.40	\$ 1,802.02	Clothing	Nike	Shorts	Good	Standard	Cash	Delivered
	9/30/2023 0:00	2023	9	#####	3	\$ 410.28	\$ 1,230.85	Grocery	Pepsi	Juice	Excellent	Unknown	Debit Card	Pending
	10/8/2023 0:00	2023	10	#####	10	\$ 195.92	\$ 1,959.17	Electronics	Samsung	Smartphone	Excellent	Express	Credit Card	Processing
	9/23/2023 0:00	2023	9	5:44:37 PM	4	\$ 15.73	\$ 62.93	Grocery	Nestle	Coffee	Bad	Same-Day	Credit Card	Pending
	10/3/2023 0:00	2023	10	1:23:14 AM	9	\$ 430.54	\$ 3,874.88	Clothing	Adidas	T-shirt	Good	Same-Day	Debit Card	Processing
	4/4/2023 0:00	2023	4	2:23:00 AM	8	\$ 337.67	\$ 2,701.38	Books	HarperCollins	Thriller	Average	Express	Debit Card	Delivered
	7/8/2023 0:00	2023	7	5:45:39 AM	6	\$ 439.84	\$ 2,639.03	Home Decor	Bed Bath & Beyond	Bedding	Average	Standard	Credit Card	Shipped
	8/20/2023 0:00	2023	8	#####	4	\$ 213.41	\$ 853.62	Grocery	Coca-Cola	Water	Excellent	Express	Debit Card	Shipped
	11/10/2023 0:00	2023	11	1:09:13 PM	10	\$ 19.38	\$ 193.83	Grocery	Pepsi	Water	Good	Express	Debit Card	Delivered
	11/25/2023 0:00	2023	11	1:08:48 PM	8	\$ 384.54	\$ 3,076.35	Grocery	Pepsi	Water	Good	Same-Day	Cash	Shipped
	5/20/2023 0:00	2023	5	3:48:46 PM	3	\$ 478.16	\$ 1,434.49	Grocery	Pepsi	Water	Good	Standard	PayPal	Delivered
	10/14/2023 0:00	2023	10	#####	1	\$ 145.98	\$ 145.98	Grocery	Pepsi	Water	Good	Same-Day	Debit Card	Delivered
	2/27/2024 0:00	2024	2	8:28:36 AM	5	\$ 96.71	\$ 483.54	Grocery	Pepsi	Water	Good	Express	Cash	Pending

Visualizations
& Insights



Dashboards



Visualizations & Insights



The Executive Summary Dashboard

Total Sales

\$138,950,147.13

Total Purchases

544,781.00

Average Rating

3.198972107

Top Country

USA

Top Category

Electronics

Total Sales

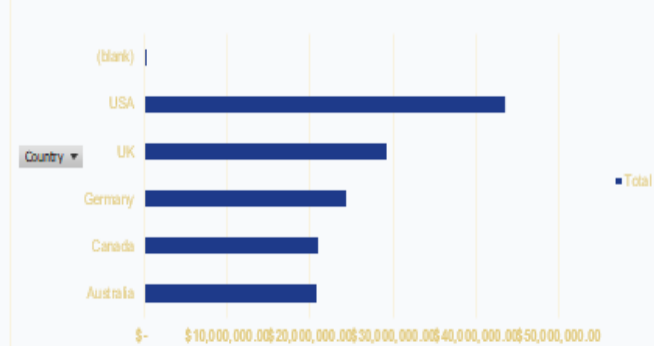
Monthly Sales Trend



Month

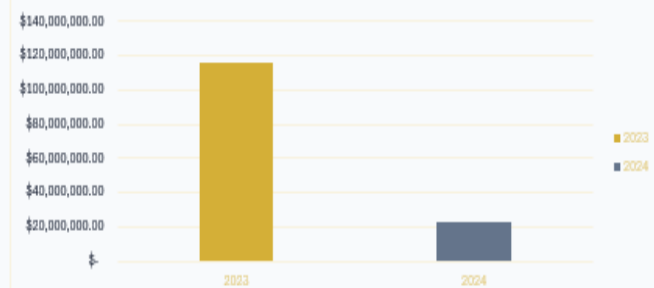
Total Sales

Sales by Country



Total Sales

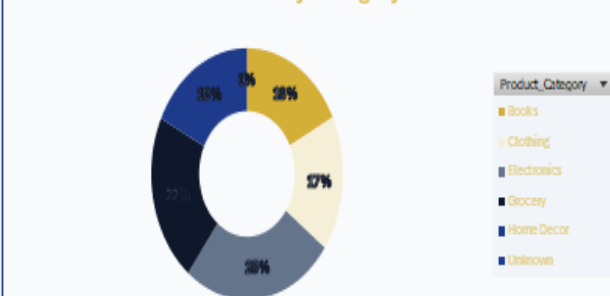
Sales by Year



Year

Total Sales

Sales by Category



Product Category

- Books
- Clothing
- Electronics
- Grocery
- Home Decor
- Unknown

Year

2023

2024

Product_C...

Books

Clothing

Electronics

Grocery

Home Decor

Unknown

Country

Australia

Canada

Germany

UK

USA

(blank)

Visualizations & Insights



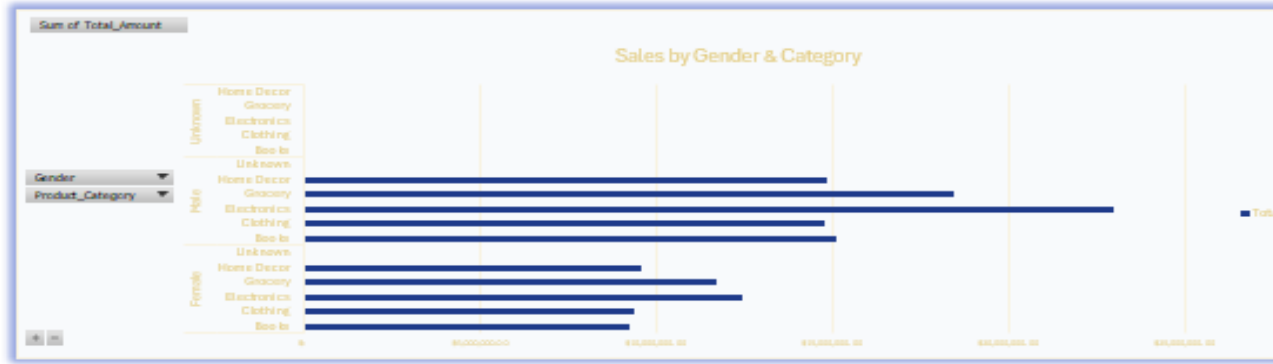
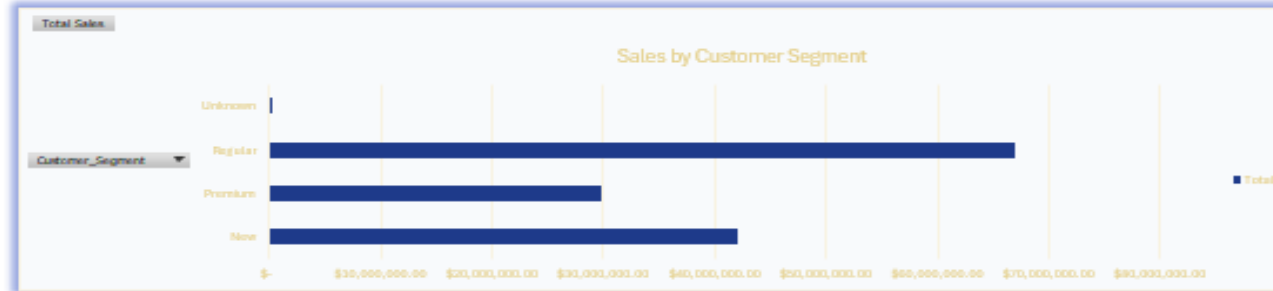
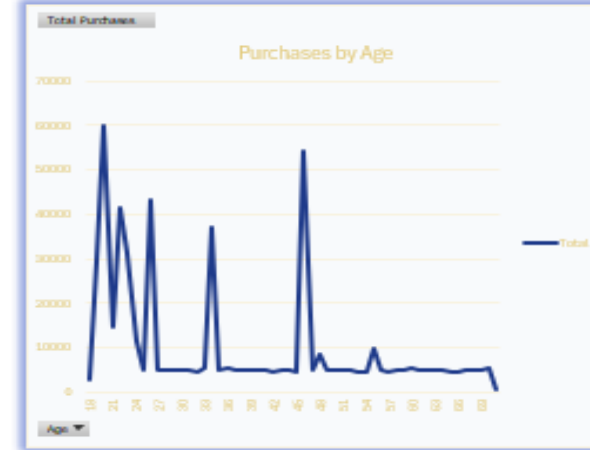
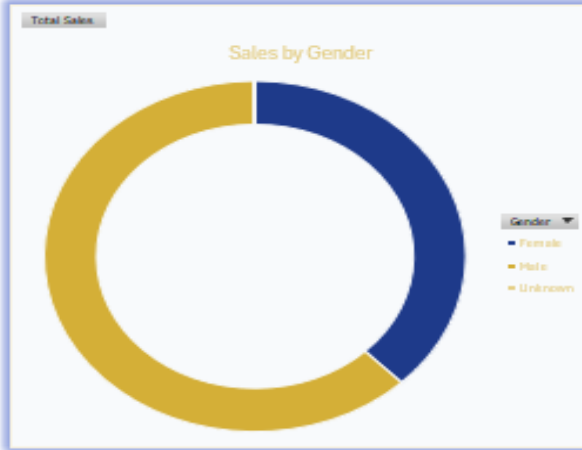
Customer Persona & Demographic DashBoard

Total Sales
\$138,950,147.13

Top Gender
Male

Top Customer Segment
Regular

Highest Purchasing Age
20



Customer

New

Premium

Regular

Unknown

(blank)

Year

2023

2024

Gender

Female

Male

Unknown

Product_C...

Books

Clothing

Electronics

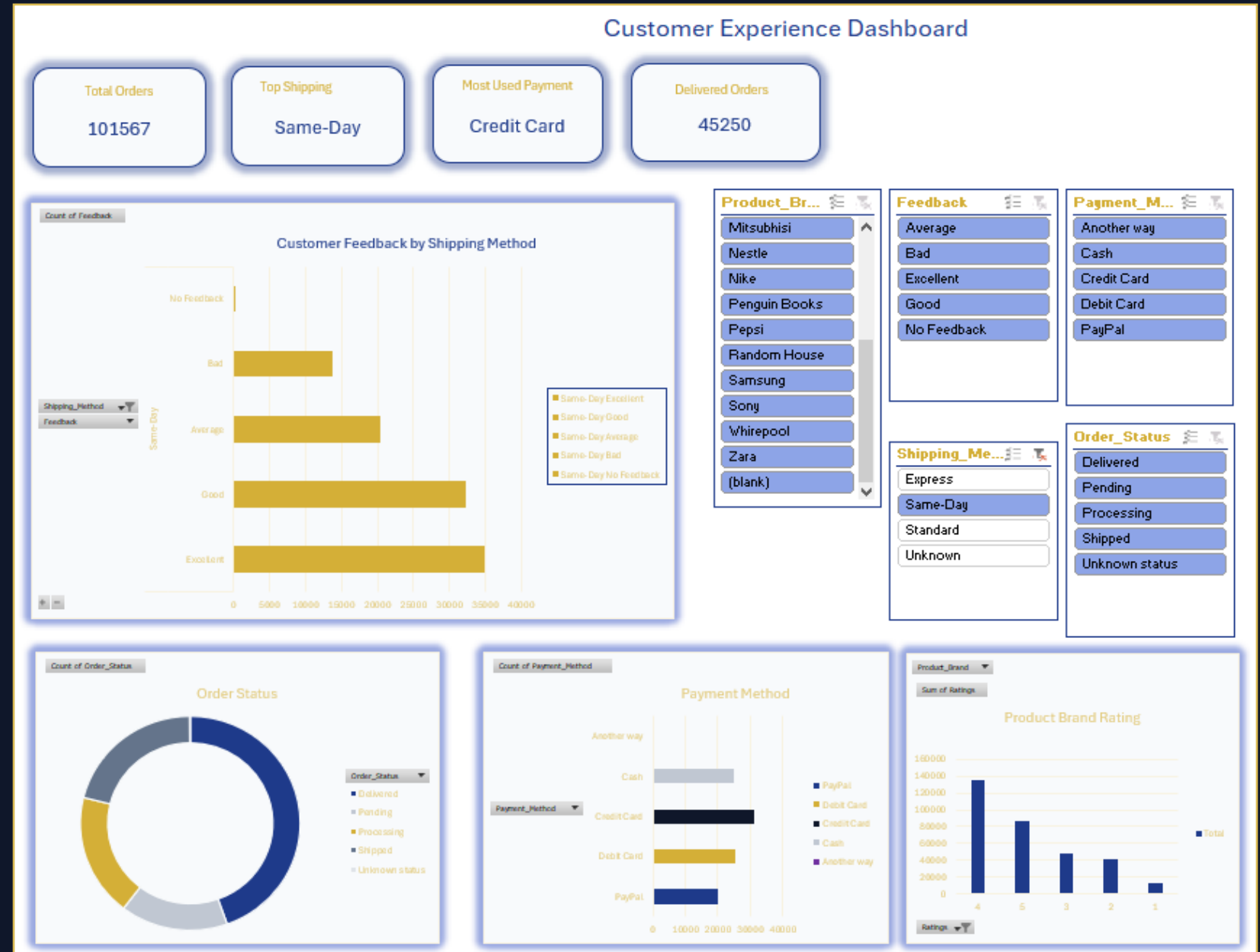
Grocery

Home Decor

Unknown



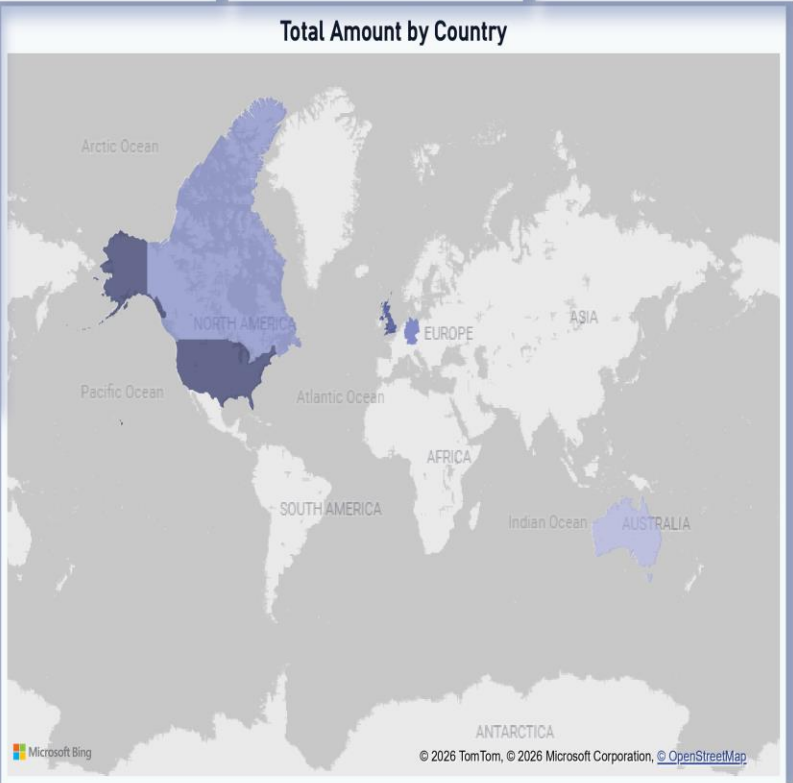
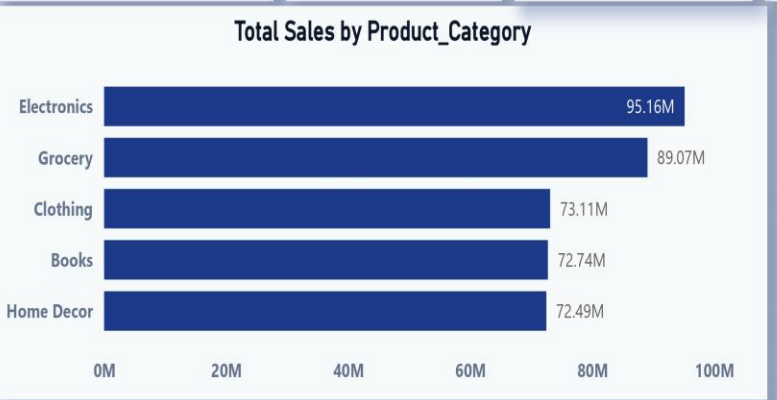
Visualizations & Insights



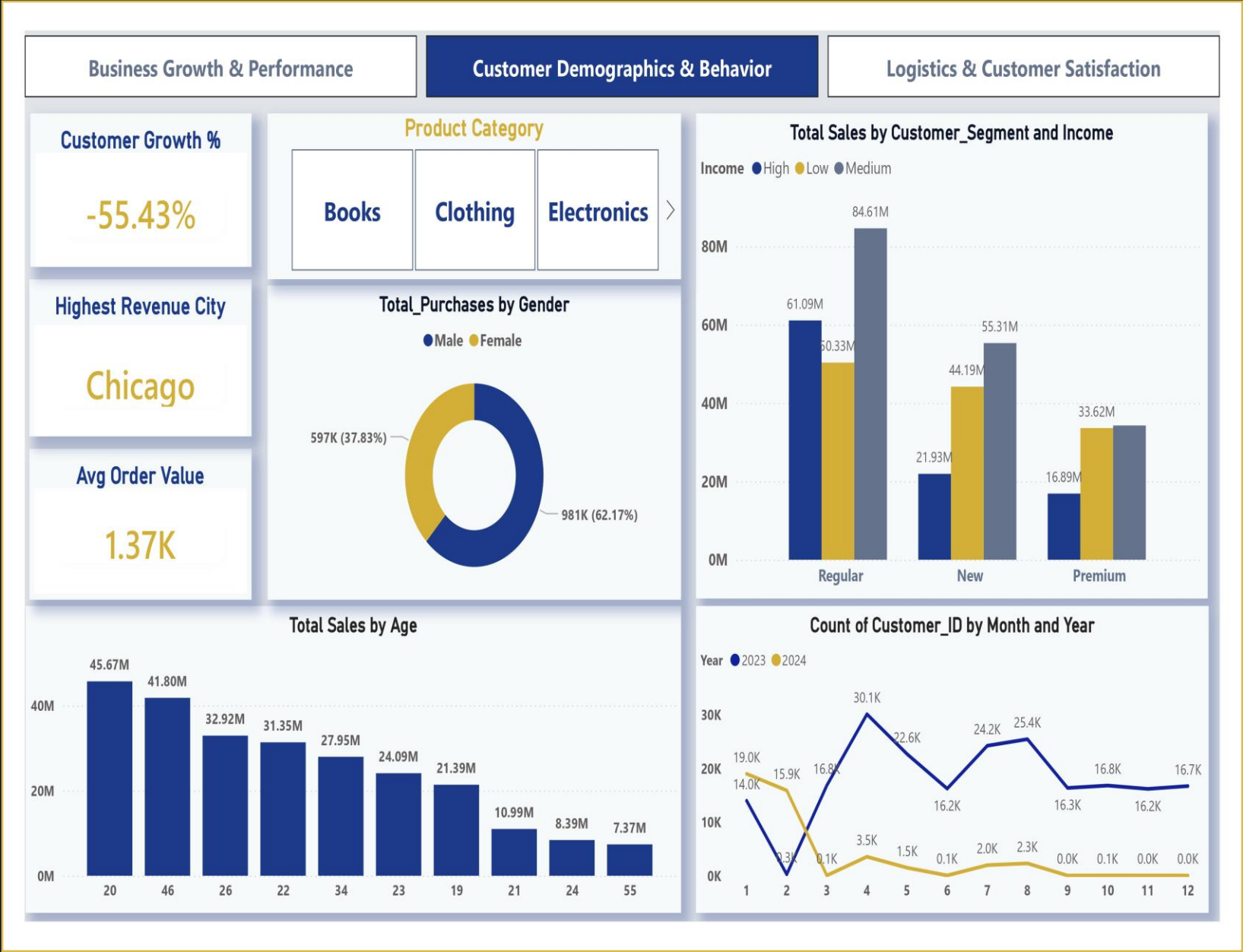


E-Commerce Retail Sales Overview

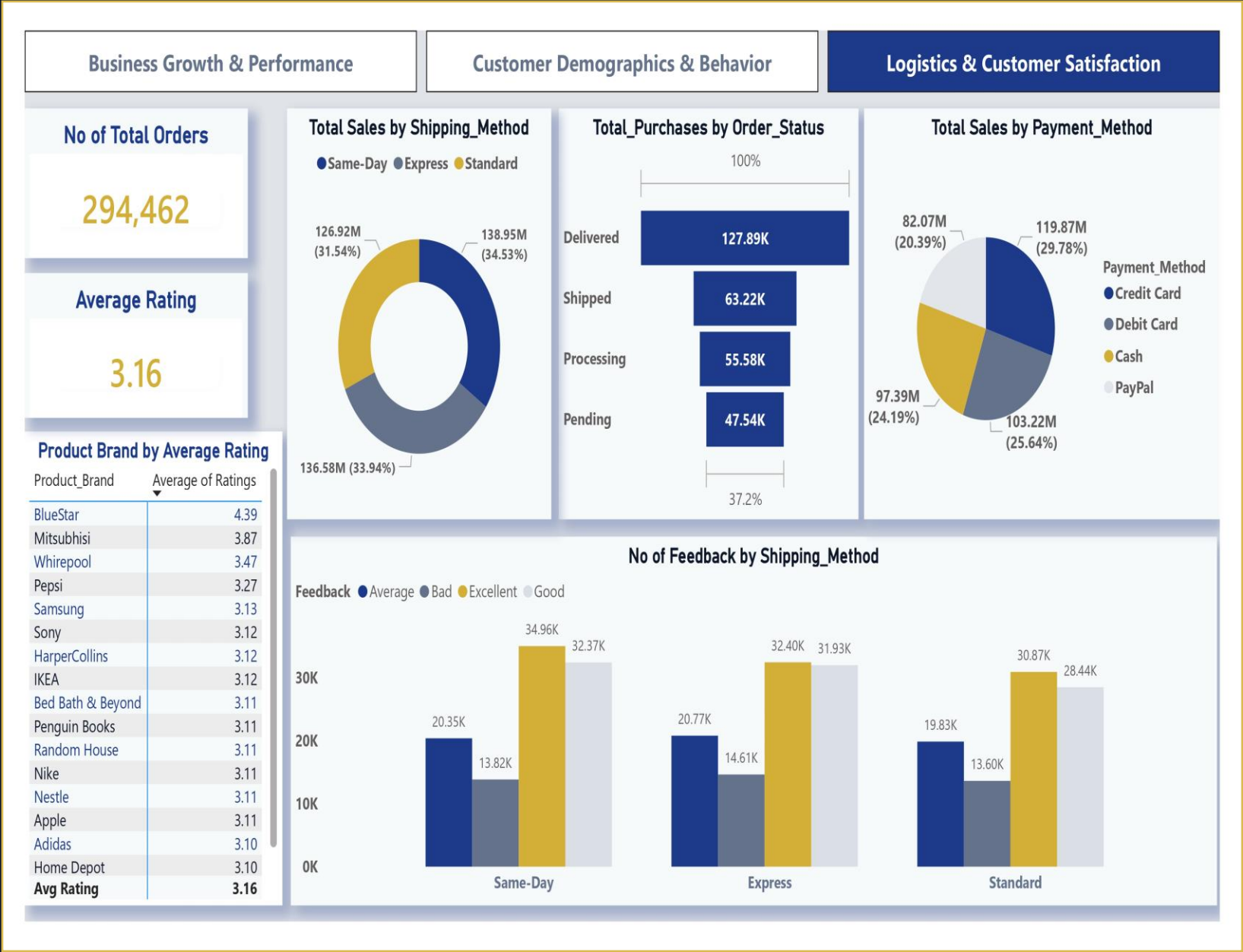
Business Growth & Performance			Customer Demographics & Behavior		Logistics & Customer Satisfaction	
Total sales	Total Purchased	Estimated Profit	Top Brand	Top Category	Year	
402.95M	2M	80.59M	Pepsi	Electronics	2023	
					2024	



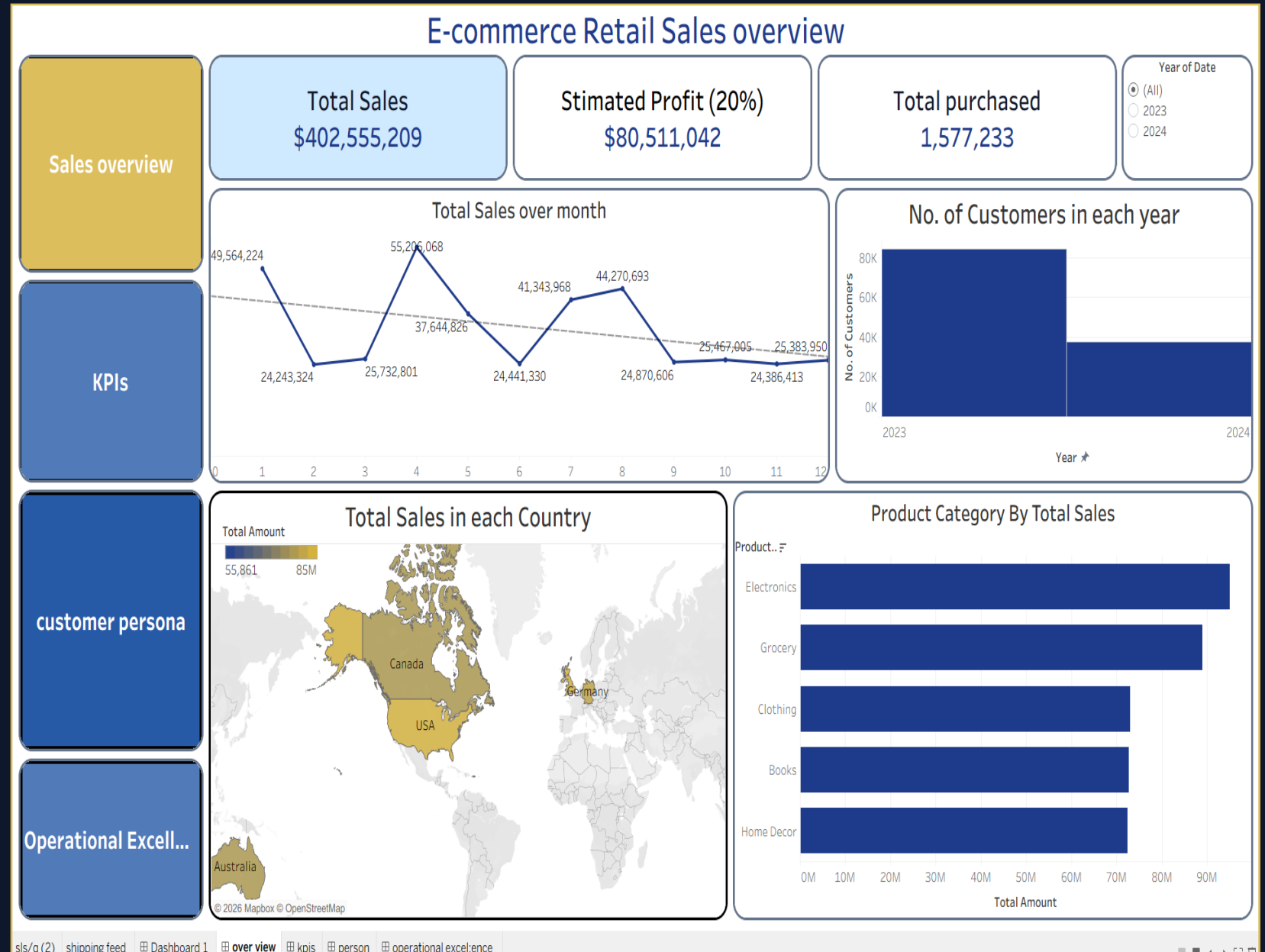
Visualizations & Insights



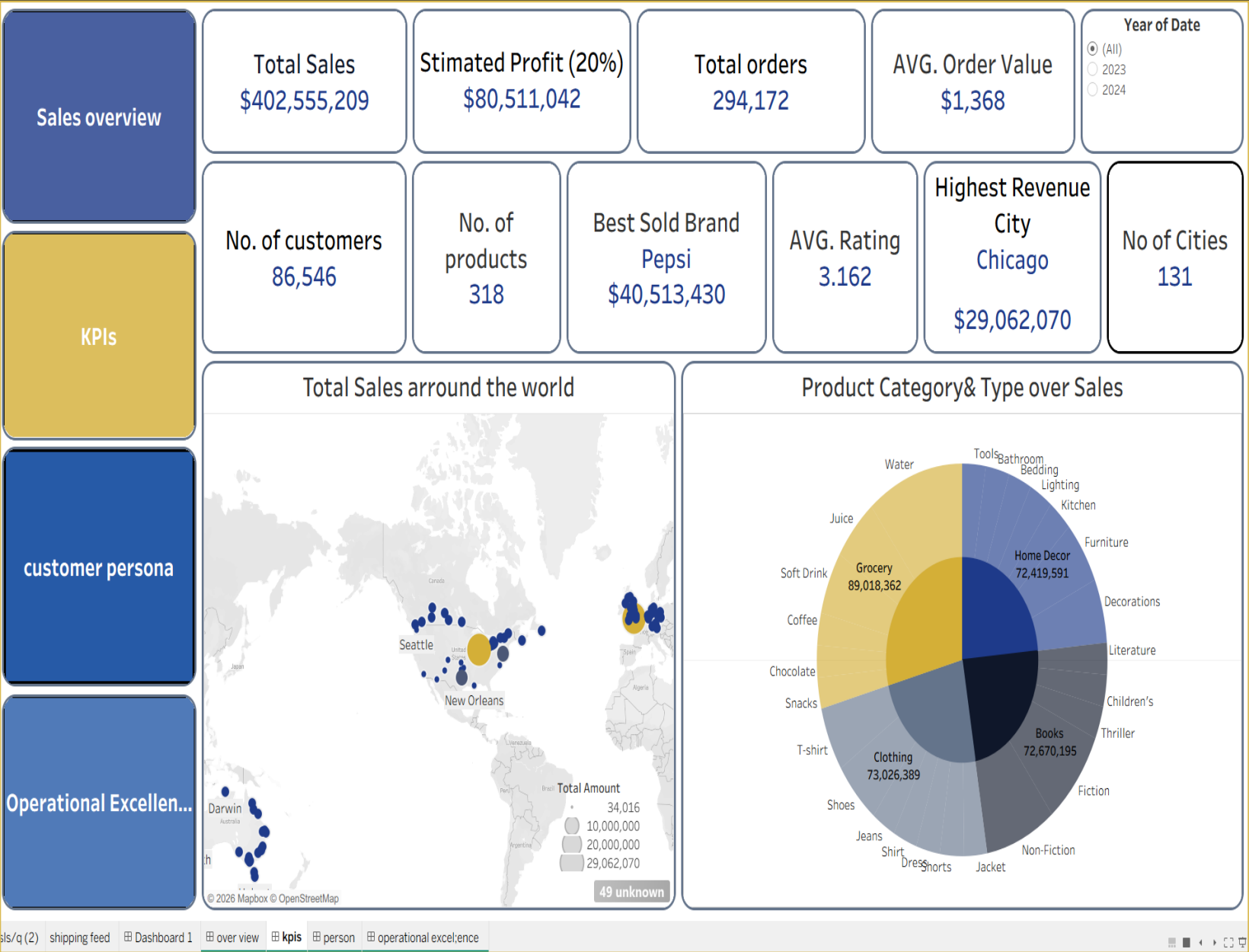
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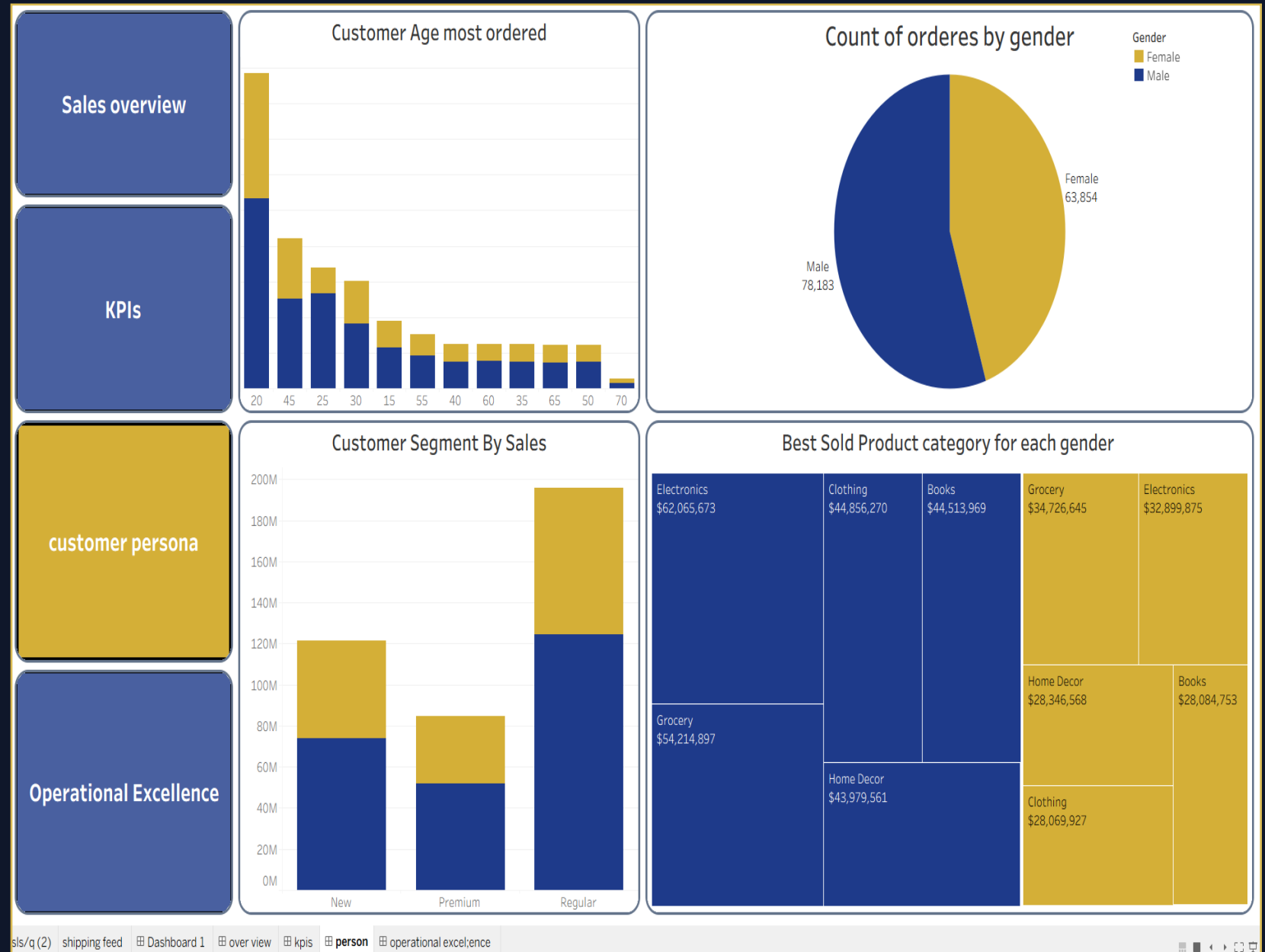
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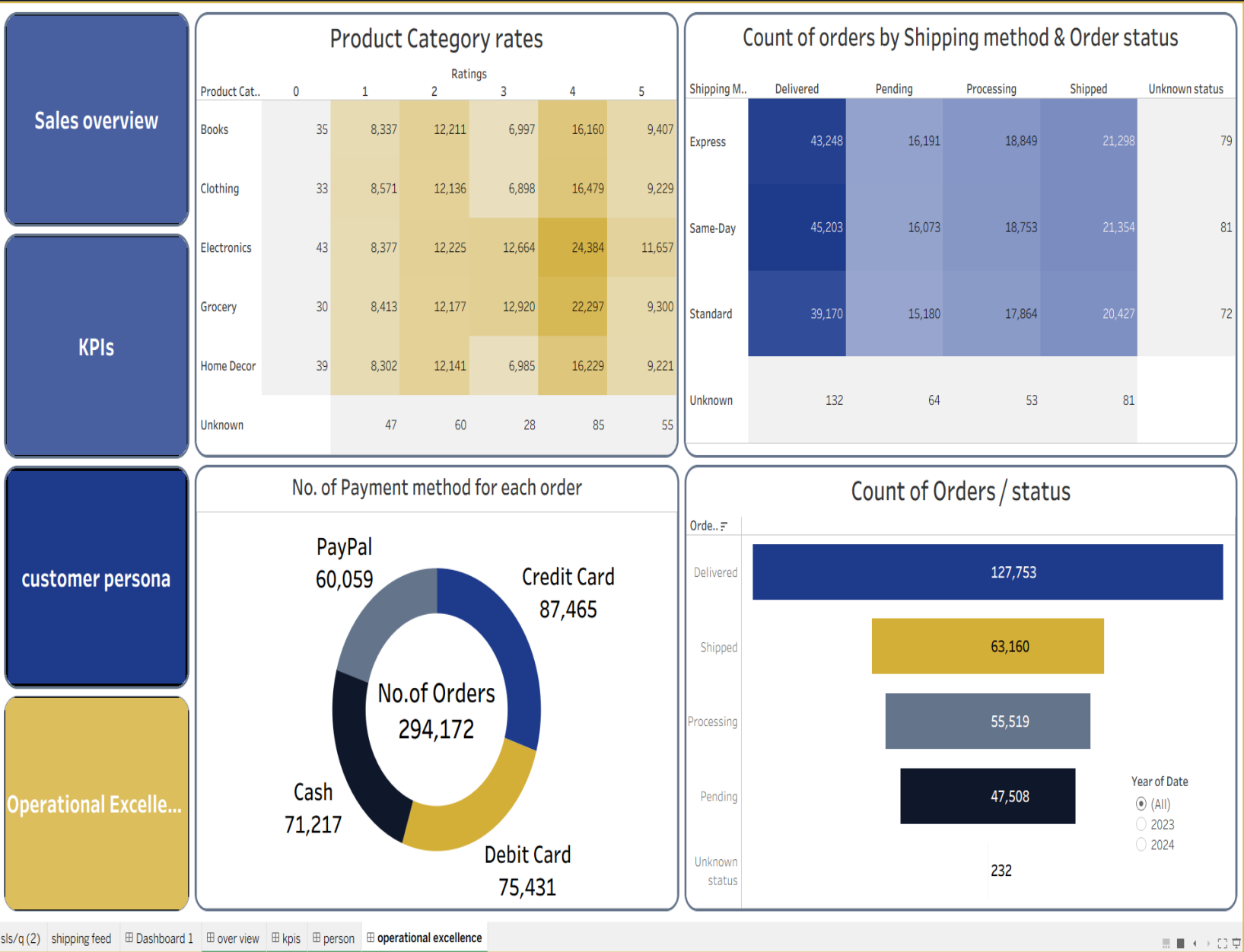
Visualizations & Insights



Visualizations & Insights



Visualizations & Insights



Insights & Recommendations



“ Based on the data insights, we are presenting 4 actionable recommendations to the business owner to secure data integrity, minimize operational bottlenecks, and maximize profitability. ”



Sales Performance & Temporal Gaps

Total Amount in:

2023
\$366M

2024
\$66M

**“ Sudden,
sharp revenue
drop in 2024
compared to
2023, causing
cash-flow
instability. ”**



Strategic Recommendations for the Business Owner

1. Data Governance & Reporting

The Advice:

Implement a **Real-Time Data Sync** and fix Customer Relationship Management (CRM) entity resolution.

The Reason: Your 2024 revenue looks like it dropped by 80%, but our technical audit proved it's just **incomplete data collection**. Fixing this prevents you from making panicked business decisions based on lagging or distorted reports.



Customer Persona & Demographic Analysis

Total Customers

Males: 78.2K

Females: 63.8K

Unknown: 310

Core Segment

Young adults aged

20 – 45

**“ Low sales from
older
demographics
>50 ”**



2. Marketing & Revenue Optimization

The Advice: Allocate **70% of your ad budget** to target young adults (aged 20–45) with Medium/High incomes in the UK and Germany.

The Reason: This demographic is your core revenue engine. Shifting away from generalized campaigns will instantly maximize your **Return on Ad Spend (ROAS)**.



Customer Persona & Demographic Analysis

| **"Regular"** segment
drives the bulk of sales.

Premium: \$85M
Regular: \$196M

“ low revenue
contribution from
the "Premium"
segment. ”



3. Customer Retention Strategy

The Advice: Launch an exclusive **VIP / Premium Loyalty Program** to convert "Regular" buyers into "Premium" ones.

The Reason: The Premium segment yields your highest **Customer Lifetime Value (CLV)**. Retaining these high-value assets is **5 times cheaper** than acquiring new ones.

Insights & Recommendations



Operational Excellence & Fulfillment

Total Orders:

294,462

Order Status:

Shipped **63.2K**

Processing **55.5K**

Delivered **127.8K**



2023 **83%**

2024 **16%**

**“ The Problem:
High backlog of
orders stuck in
Shipped,
Processing, and
Pending statuses,
indicating
shipping
bottlenecks. ”**



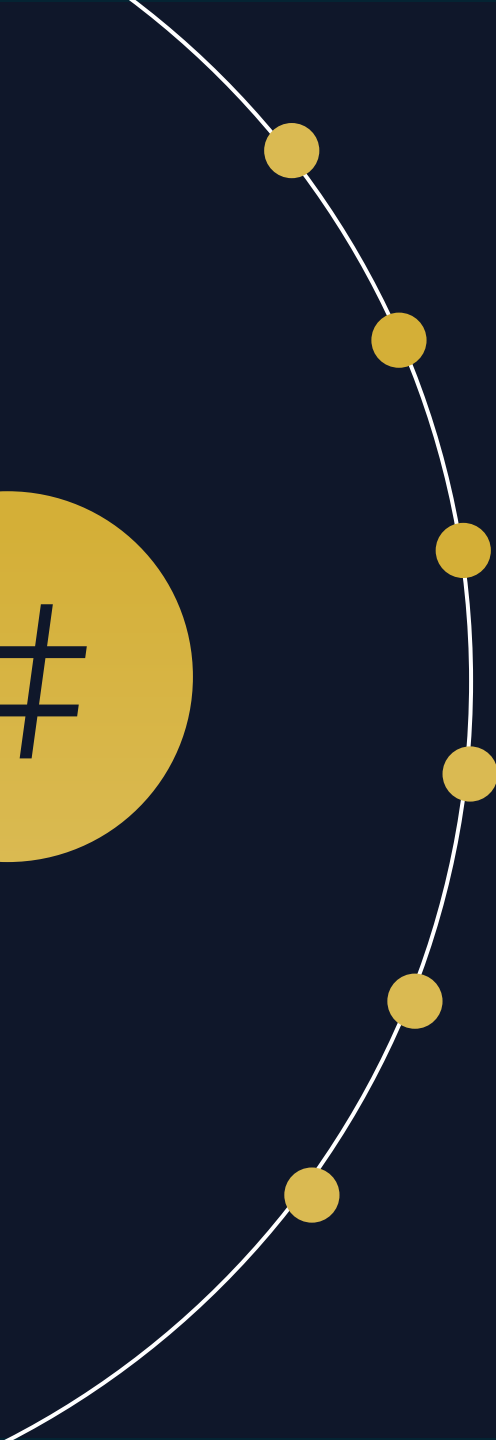
4. Supply Chain & Logistics Overhaul

The Advice: Audit and review current shipping vendors to accelerate order fulfillment.

The Reason: A massive backlog of orders is currently stuck in "**Shipped**" and "**Processing**" statuses. Clearing this bottleneck will reduce order cancellations and boost customer satisfaction.

Questions ?

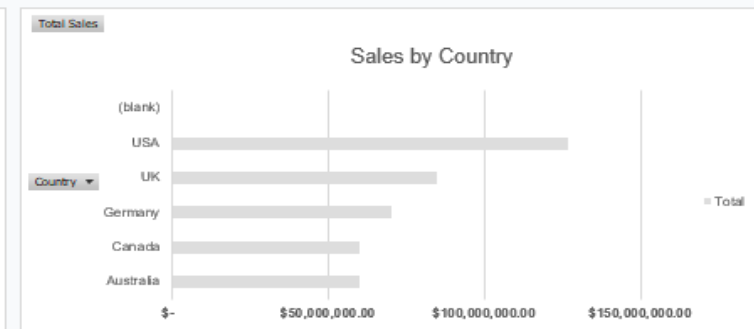




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THE END

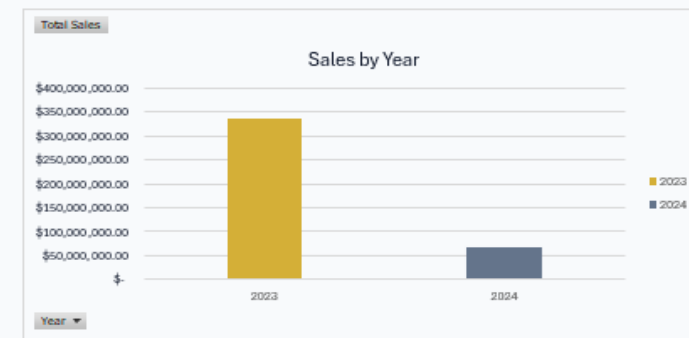
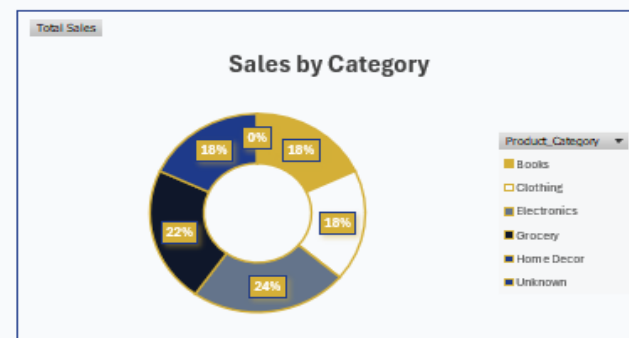
Backstage



Year

2023

2024



Product_Ca...

Books

Clothing

Electronics

Grocery

Home Decor

Unknown

Country

Australia

Canada

Germany

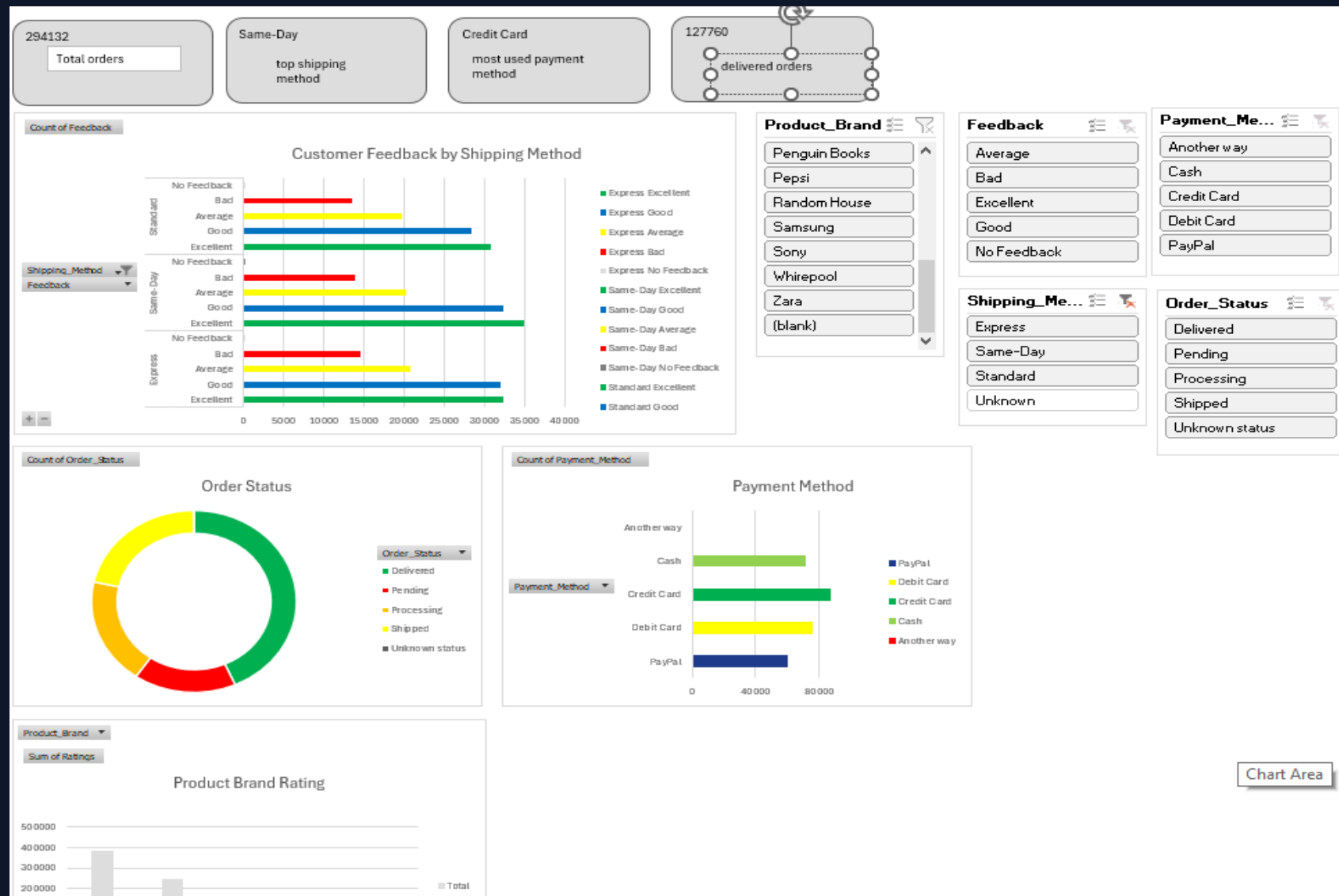
UK

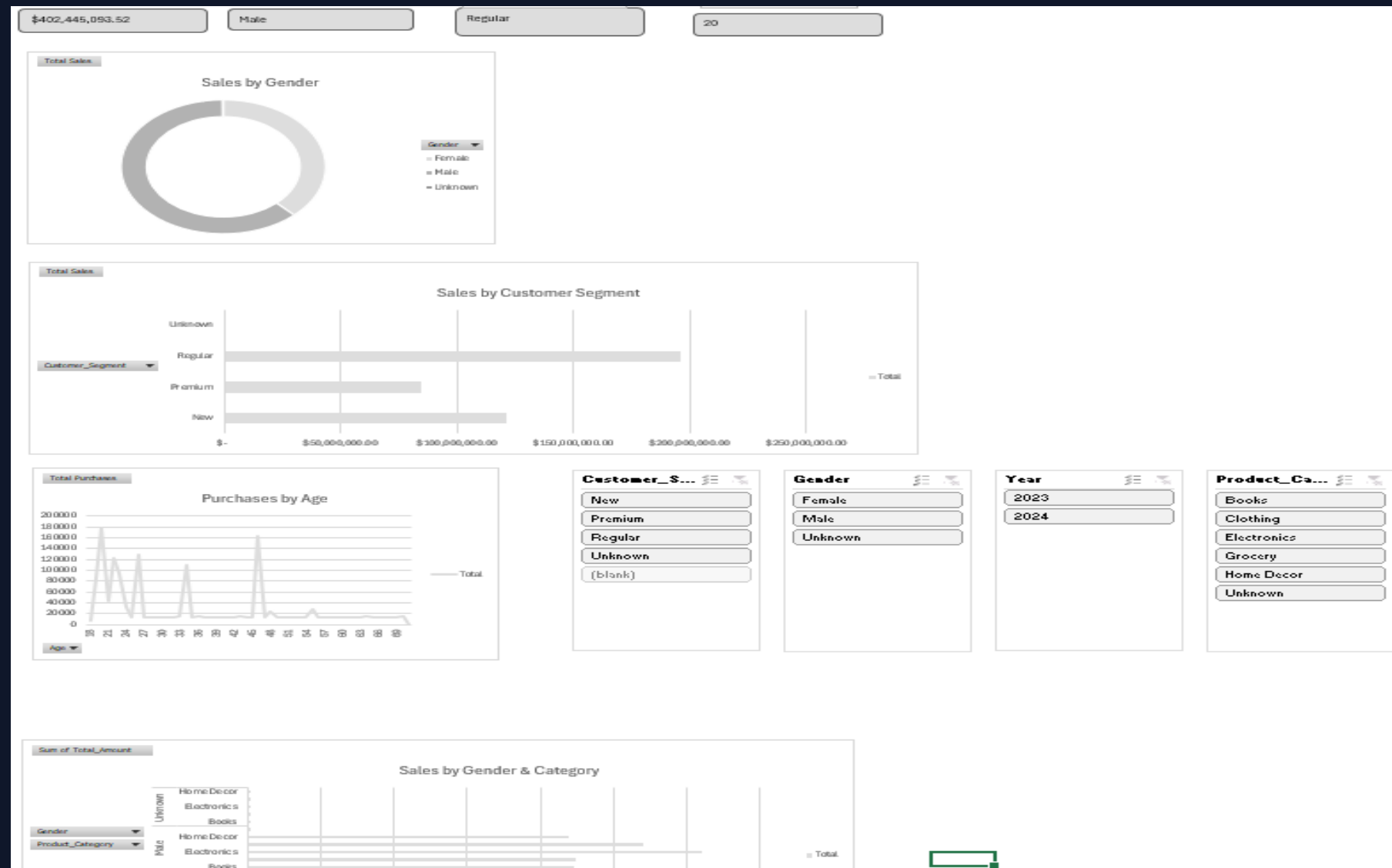
USA

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Executive Insights

USA contributes the highest share of sales.
Electronics is the top-performing category.







File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Show Me

Data Analytics Pages Columns Rows

first version of data (final ...)

Search

Tables

- Phone
- Product hier
 - Product Category
 - Product Type
 - Product Brand
 - Products
- Ratings
- Shipping Method
- State
- Time
- Transaction ID
- Year
- Zipcode
- Measure Names
- Amount
- avg order value
- Mkt(-1)
- Estimated Profit (20%)
- Total Amount
- Total Purchases
- first version of data (Con...
- Latitude (generated)
- Longitude (generated)
- Measure Values

Parameters

- AxisEnd

Filters

- Payment Method
- YEAR(Date)
- Action (CO- Co-)
- Action (Product-)

Marks

Automatic

Color Size Text

Detail Tooltip

ADD avg order ...

avg order value

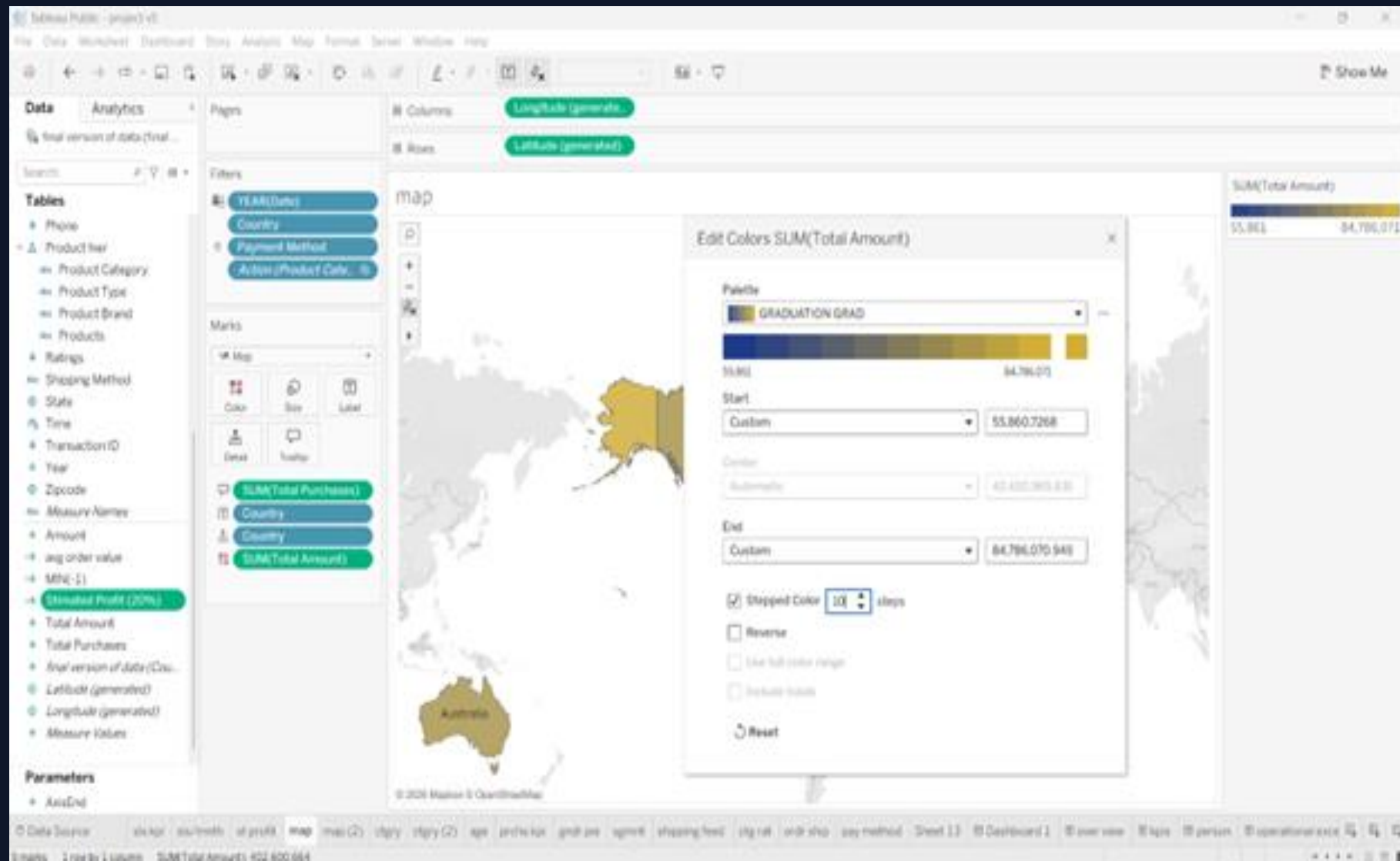
`SUM([Total Amount])/COUNT([Transaction ID])`

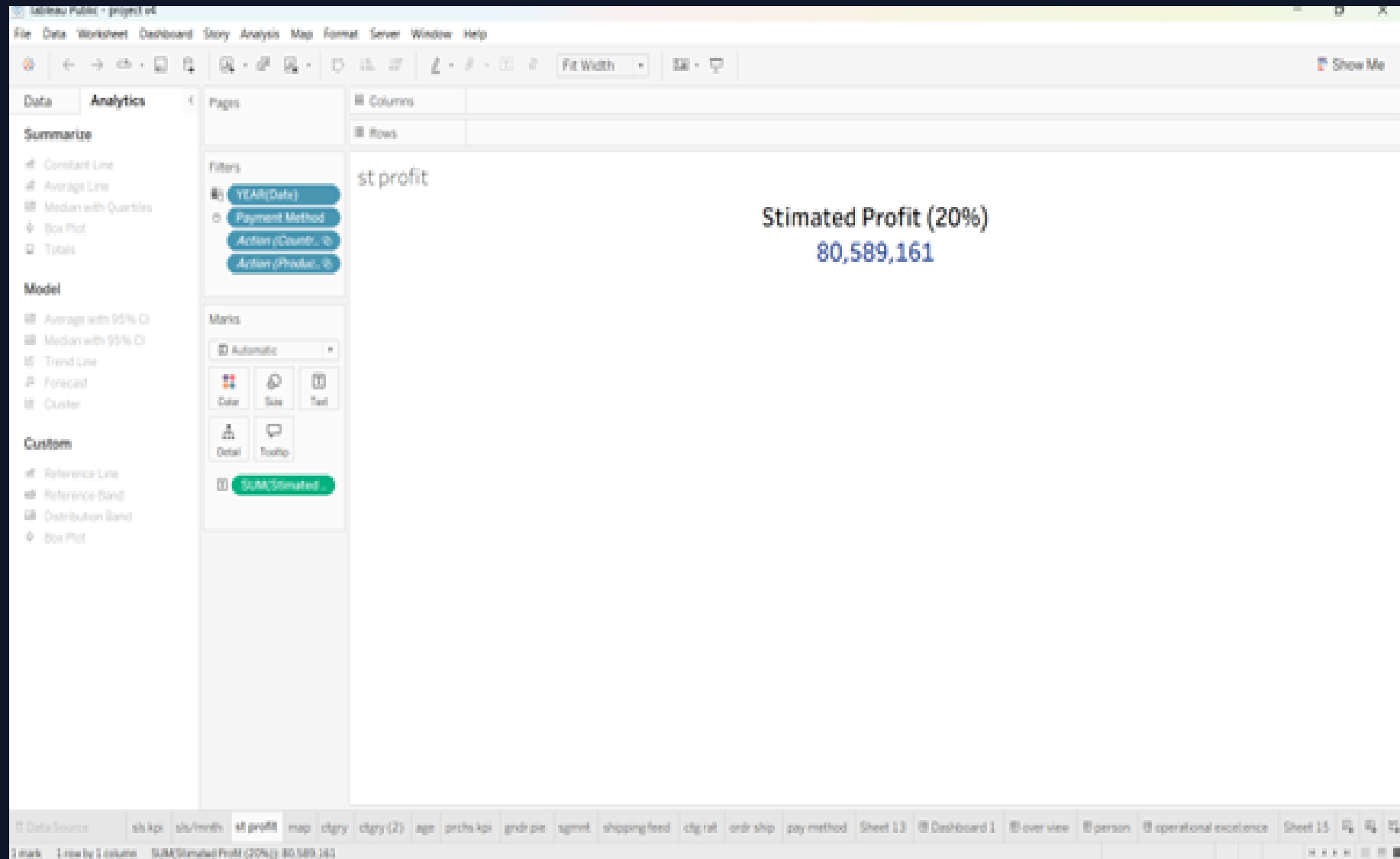
The calculation is valid. 2 Dependencies

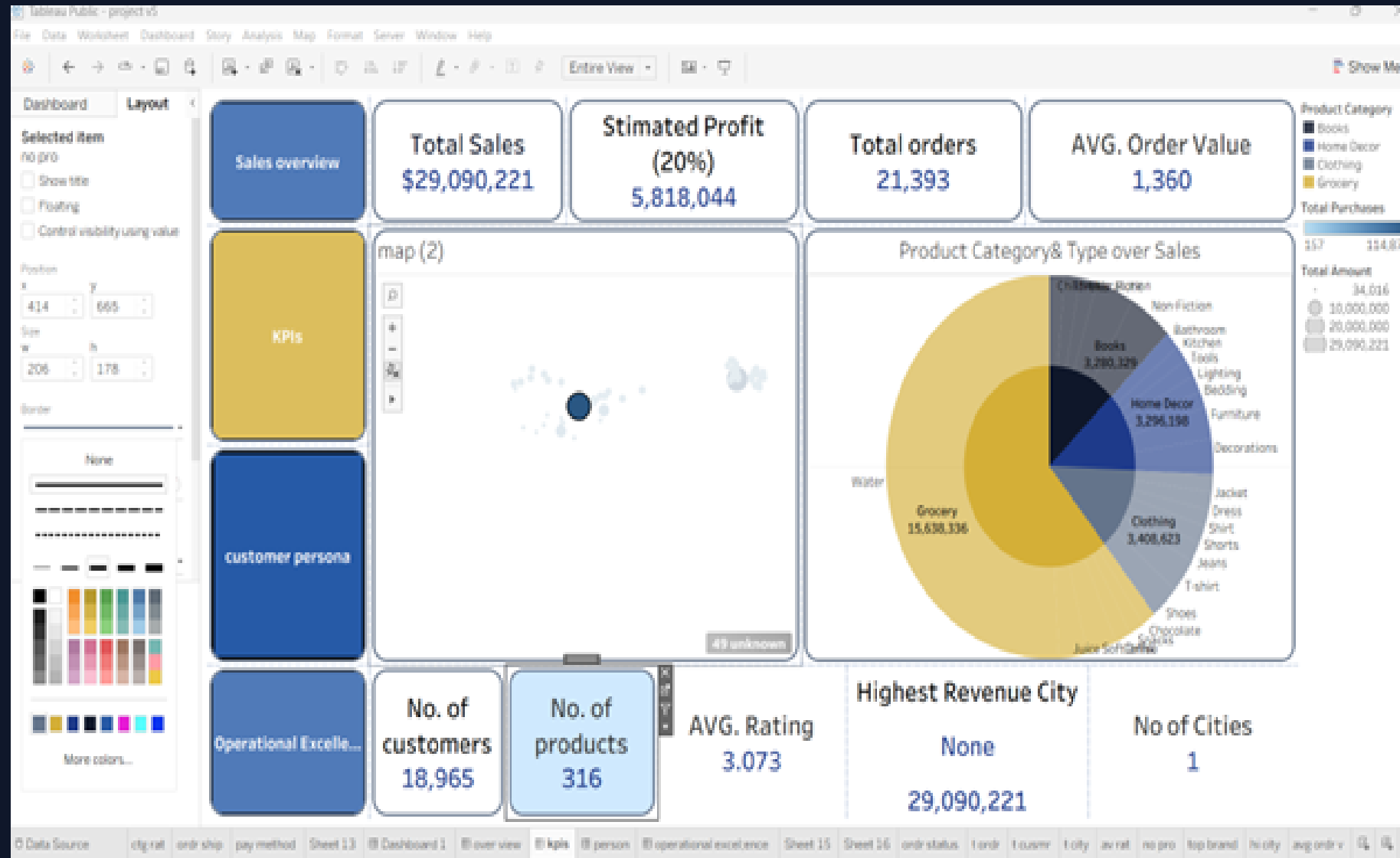
Cancel OK

Data Source: avg order value, shipping, payment method, Sheet 13, Dashboard 1, Overview, Sign, Person, Operational Excellence, Sheet 15, Sheet 16, web status, first version, fully, web, recipe, top brand, fully, avg order value

mark 1 row by 1 column SUM of Adding order value \$1,388







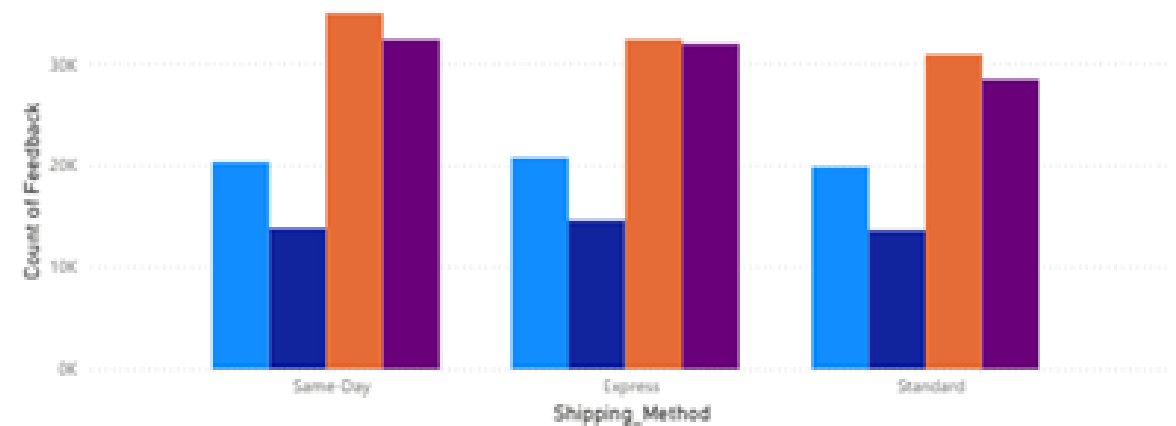




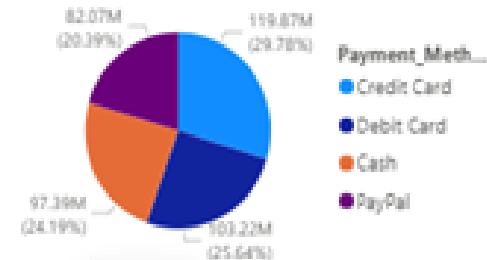
Product_Brand	Average of Ratings
BlueStar	4.39
Mitsubishi	3.87
Whirepool	3.47
Pepsi	3.27
Samsung	3.13
Sony	3.12
HarperCollins	3.12
IKEA	3.12
Bed Bath & Beyond	3.11
Penguin Books	3.11
Random House	3.11
Nike	3.11
Nestle	3.11
Apple	3.11
Adidas	3.10
Home Depot	3.10
Coca-Cola	3.10
Zara	3.10
Total	3.16

Count of Feedback by Shipping_Method and Feedback

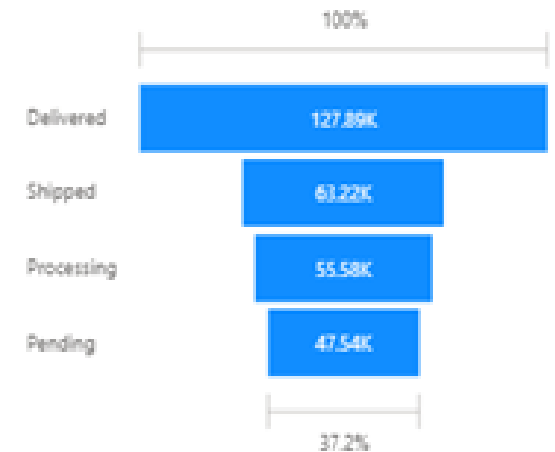
Feedback: Average (Blue), Bad (Dark Blue), Excellent (Orange), Good (Purple)



Sum of Total_Amount by Payment_Method



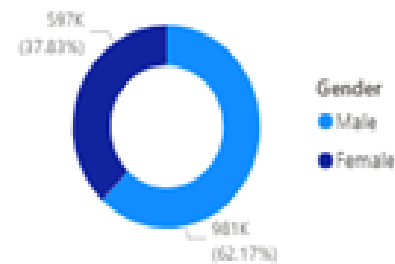
Count of Total_Purchases by Order_Status



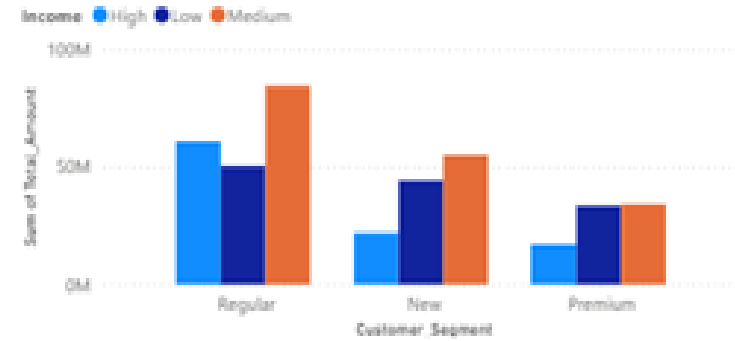
Backstage



Sum of Total_Purchases by Gender



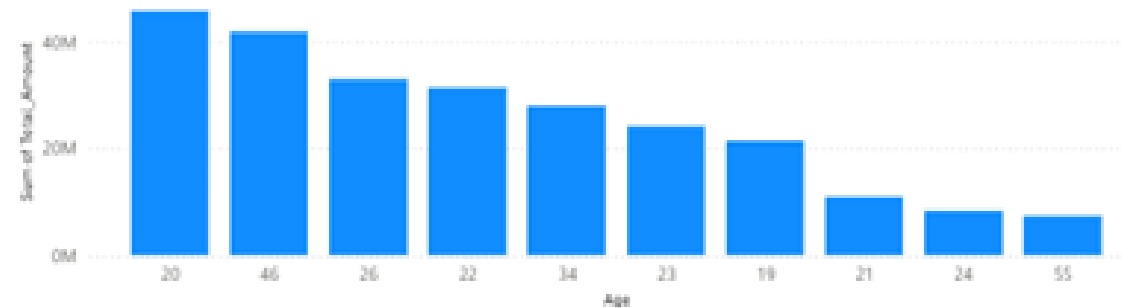
Sum of Total_Amount by Customer_Segment and Income



Product_Category

- ☐ Books
- ☐ Clothing
- ☐ Electronics
- ☐ Grocery
- ☐ Home Decor

Sum of Total_Amount by Age





```
1 Estimated Profit = SUM('final version of data'[Total_Amount]) * 0.20
```



```
1 Highest Revenue City =  
2 MAXX(  
3     TOPN(  
4         1,  
5         VALUES('final version of data'[City]),  
6         CALCULATE(SUM('final version of data'[Total_Amount]))  
7     ),  
8     'final version of data'[City]  
9 )
```



```
1 Customer Growth % =  
2 VAR Customers2023 = CALCULATE(DISTINCTCOUNT('final version of data'[Customer_ID]), 'final version of data'[Year] = 2023)  
3 VAR Customers2024 = CALCULATE(DISTINCTCOUNT('final version of data'[Customer_ID]), 'final version of data'[Year] = 2024)  
4 RETURN  
5 DIVIDE(Customers2024 - Customers2023, Customers2023)
```



```
1 avg order value = DIVIDE(SUM('final version of data'[Total_Amount]),COUNT('final version of data'[Order_Status]))
```